



Cloud Operating Systems and OS- Level Virtualization

Course code & Name : CSA0406 – OPERATING SYSTEMS

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What is Cloud Operating System ?

Managing Computing Resources Anytime, Anywhere

Library	Cloud Operating System
Stores many books	Manages many computing resources
Organizes books into sections	Organizes servers, storage, and networks
Issues books to readers	Allocates resources to users
Tracks borrowed books	Monitors resource usage
Serves many readers at once	Supports multiple users simultaneously



Just as a library manages books and provides them to many readers efficiently, a Cloud Operating System manages computing resources and delivers them to users over the internet whenever needed.

PROBLEM IDENTIFICATION

- High hardware and maintenance costs.
- Limited storage and computing resources.
- Difficult to scale resources as user demand increases.

This leads to inefficient resource utilization, slower performance, and increased operational costs.

Where Does the problem Occur ?

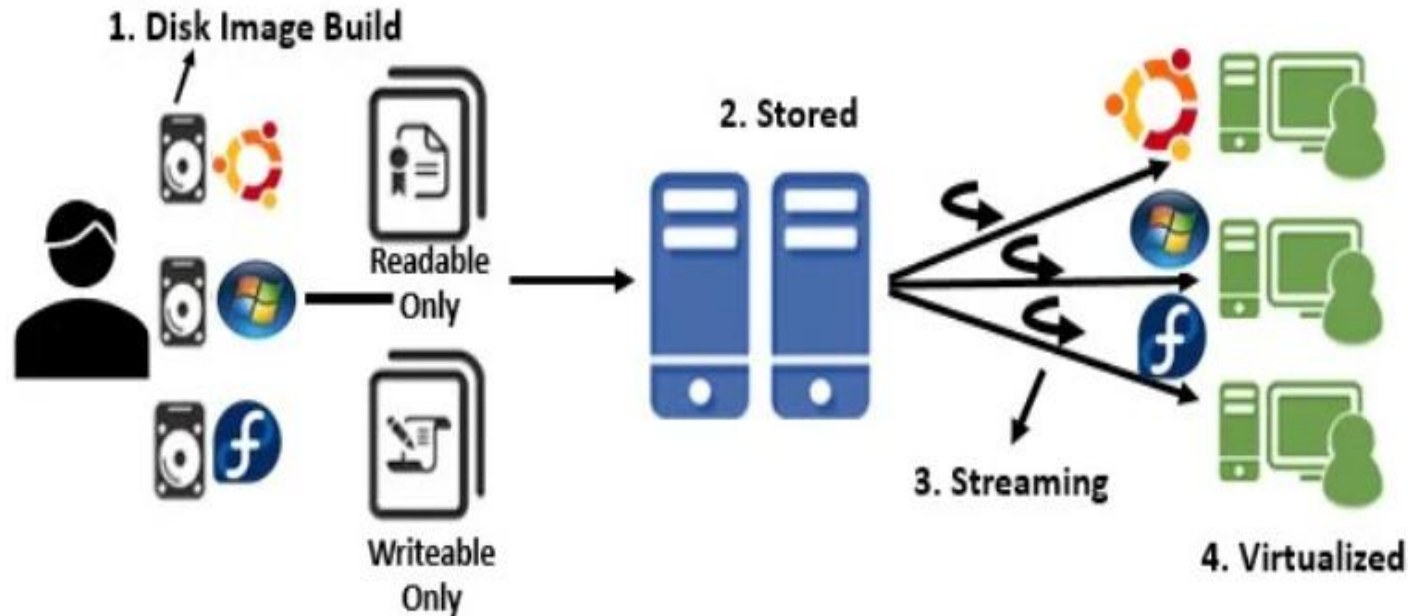
- Organizations with growing user demands.
- Data centers managing multiple applications.
- Businesses requiring remote access and high availability.



OS-Level Virtualization (Architecture)

What is OS-Level Virtualization?

OS-Level Virtualization allows multiple isolated environments called **containers** to run on a single operating system kernel.

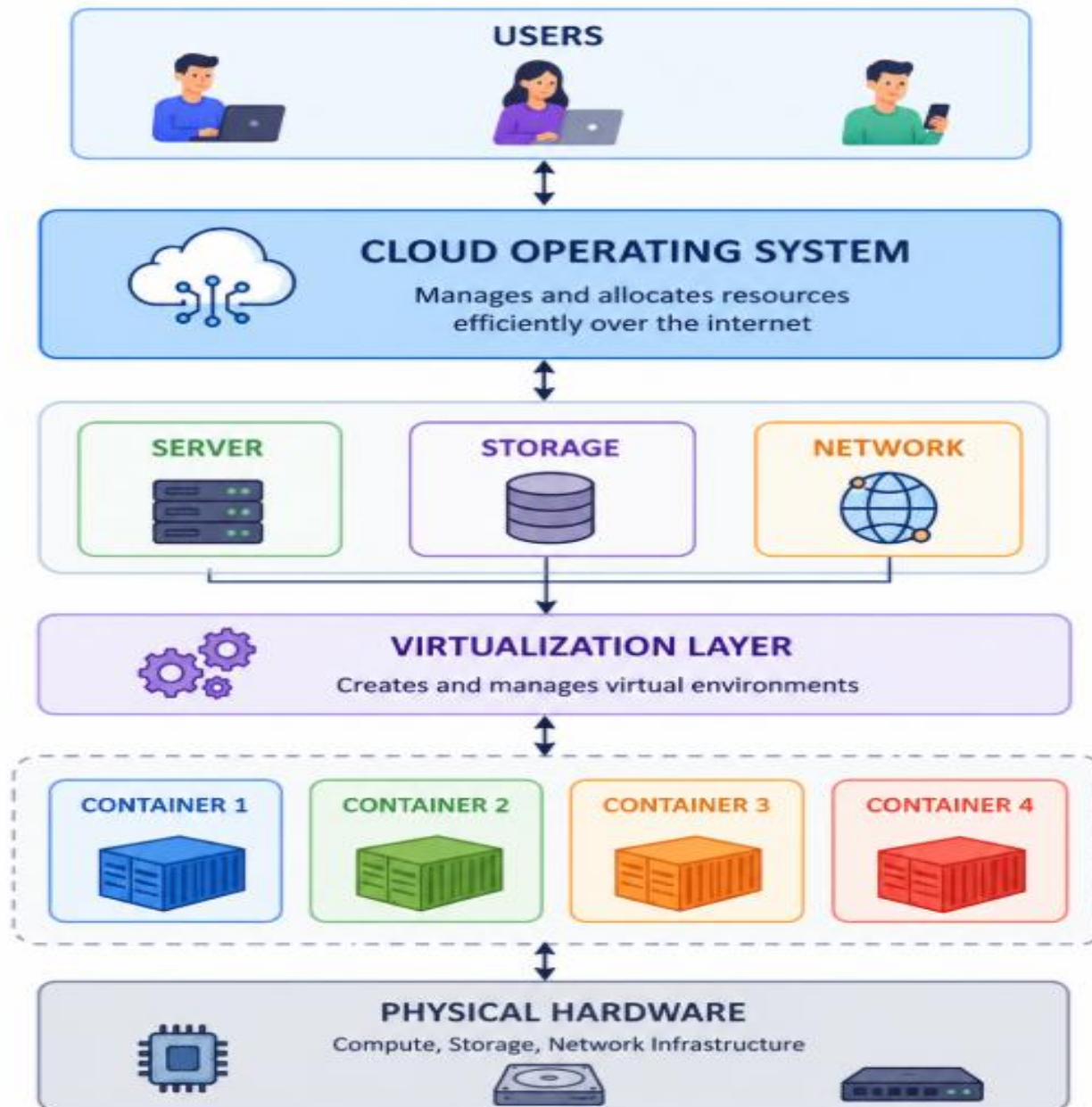


HOW IT WORKS

- 1** The OS kernel is shared by all containers.
- 2** Each container has its own user space with applications, libraries and dependencies.
- 3** Containers are isolated from each other but share the same kernel.



Engineer to Excel



Benefits

- Fast deployment
- Lightweight containers
- Better resource utilization
- High performance

COMPARISON

Traditional OS	Cloud OS
Installed on one computer.	Accessible through the internet
Limited resources.	Scalable resources
Manual maintenance	Centralized management
Higher hardware cost.	Pay-as-you-use model
Less flexible	Highly flexible



Engineer to Excel



ADVANTAGES OF CLOUD OPERATING SYSTEMS



Lower Infrastructure Cost

Pay only for what you use and reduce hardware expenses.



Easy Scalability

Scale resources up or down based on demand.



High Availability

Ensure continuous service with minimal downtime.



Faster Application Deployment

Deploy applications quickly and bring ideas to life faster.



Efficient Resource Sharing

Share resources efficiently across multiple users.



Better Disaster Recovery

Recover data and services quickly in case of failures.

REAL-WORLD APPLICATIONS



Google Drive

Stores files securely in the cloud and access from anywhere.



Gmail

Provides email services with high availability and storage.



Netflix

Streams content to millions of users worldwide with high performance.



YouTube

Delivers videos seamlessly to billions of users across the globe.



Microsoft Azure

Offers cloud services like computing, storage, and databases.



Amazon Web Services (AWS)

Provides a wide range of scalable cloud services worldwide.



Cloud Operating Systems help organizations save costs, scale easily, stay available, and deliver better services to users around the world.





CONCLUSION

- ❖ Cloud Operating Systems simplify the management of computing resources.
- ❖ OS-Level Virtualization improves efficiency by running multiple containers on one operating system.
- ❖ These technologies are widely used in modern cloud computing because they provide scalability, flexibility, and cost savings.

Thank You!